OSPF

From Dynamic Routing Fundamentals to Configuration

Study Notes

Contents

1 Classification of Dynamic Routing Protocols

1.1 By Application Scope (AS)

- **Interior Gateway Protocols (IGPs)** – run *inside* an Autonomous System: **RIP**, **OSPF**, **IS-IS**
- **Exterior Gateway Protocols (EGPs)** – run *between* Autonomous Systems: **BGP**

1.2 By Working Mechanism / Algorithm

- **Distance-vector routing protocols**: **RIP**
- **Link-state routing protocols**: **OSPF**, **IS-IS**
- **BGP** uses the **path-vector algorithm** – a modified version of the distance-vector algorithm.

1.3 Distance-Vector Routing Protocols

A router running a distance-vector protocol **periodically floods** its routing table to neighbors through route exchange. Each router **learns routes from its neighbors**, loads them into its own routing table, and **re-advertises** them onward.

Key limitation: routers running distance-vector protocols do *not* know the overall network topology – they only know the **direction** (next hop) and **cost** to a destination.

1.4 Link-State Routing Protocols

A link-state protocol advertises **link state information** – not routing table information directly.

1. Routers establish **neighbor relationships**.
2. Routers exchange **Link State Advertisements (LSAs)** – descriptions of link status, not routes.

Each router generates its own LSA describing its **directly connected interfaces only**: interface cost and the relationship with neighboring routers.

LSDB maintenance: each router stores its own LSAs *and* the LSAs received from others in its **LSDB (Link State Database)**. Routers parse the LSDB to reconstruct the **entire network topology** – fundamentally different from distance-vector, where no router ever builds a full topology map.

1.5 Quick Comparison

	Distance-Vector (RIP)	**Link-State (OSPF/IS-IS)**
What's exchanged	Full routing table	Link state info (LSAs)
Knowledge of topology	No – only direction + cost	Yes – full topology from LSDB
Update behavior	Periodic full flooding	Neighbor relationships + LSA flooding
Algorithm	Bellman-Ford	Dijkstra / SPF

2 SPF Calculation

Every router uses the **Shortest Path First (SPF)** algorithm to calculate routes, based on its **LSDB**. Each router calculates a **loop-free tree**, with **itself as the root**, representing the shortest path to every other node – giving it the optimal path to every network segment.

SPF is the core algorithm of OSPF – it enables OSPF to select optimal routes even on complex networks.

Routing table generation: once SPF calculates the optimal path tree, the router installs those best paths into its **routing table (RIB)**.

2.1 The 4 Steps of a Link-State Protocol

Step	Action
1	Establish **neighbor relationships** between neighboring routers
2	**Exchange link state information** (LSAs) and **synchronize LSDBs** between neighbors
3	**Calculate the optimal path** – run SPF on the LSDB to build the shortest-path tree
4	**Generate routing entries** from the shortest-path tree and load them into the routing table

Key insight: because every router in the same area has an *identical* LSDB (Step 2), but each router runs SPF independently with itself as root (Step 3), every router ends up with a consistent, loop-free view of the network – just seen from its own position.

3 Overview of OSPF

OSPF, defined by the **IETF**, is an **IGP based on link state**.

- **OSPFv2** (RFC 2328) – for **IPv4**
- **OSPFv3** (RFC 2740) – for **IPv6**

3.1 OSPF Advantages

1. Uses **accumulated link cost** as the reference value for route selection, based on the **SPF algorithm**.
2. Transmits and receives certain protocol packets in **multicast mode**.
3. Supports **area partition** – dividing a large network into smaller, manageable areas.
4. Supports **load balancing** among equal-cost routes.
5. Supports **packet authentication**.

4 OSPF Application Scenarios

OSPF is typically deployed on large-scale enterprise networks, structured in three layers.

4.1 The Three-Layer Model

Layer	Role
Access	Connects end users/terminals to the network via fiber, twisted pair, coax, or wireless. Low-cost, high port density devices.
Aggregation	Provides policy-based connections between access and core – address aggregation, protocol filtering, routing, authentication. Divides network segments to isolate faults and prevent them from spreading to the core.
Core	Optimized transmission between backbone networks – focused on redundancy, reliability, high-speed transmission.

How OSPF areas map to these layers:

- **Core + Aggregation layers** → deployed in the **OSPF backbone area** (Area 0).
- **Access + Aggregation layers** → deployed in **OSPF non-backbone areas** (Area 1, Area N, . . .).

5 Basic OSPF Concepts

5.1 Router ID

A **Router ID** is a **32-bit integer** that uniquely identifies an OSPF router within an AS.

Selection rules (in priority order):

1. **Manually configured** – recommended.
2. If not configured → the **largest IP address among loopback interfaces**.
3. If no loopback exists → the **largest IP address among physical interfaces**.

Practical notes:

- To **change** an active router ID, the OSPF process must be **restarted**.
- **Best practice:** plan a private segment (e.g. `192.168.1.0/24`) for router ID selection; before starting OSPF, create a loopback interface on each router with a private /32 address, and use that as the router ID. This loopback address generally does not need to be advertised into OSPF.
- **Why a loopback?** Loopback interfaces never go physically down, so the router ID stays stable even if physical interfaces fail.

5.2 Area

Each OSPF area is a **logical group**, identified by an **Area ID** – a 32-bit non-negative integer, written in dotted-decimal notation (same format as an IPv4 address), e.g. `area 0.0.0.1`. It can also be expressed in plain decimal.

Dotted-decimal	Decimal equivalent
`area 0.0.0.1`	`area 1`
`area 0.0.0.255`	`area 255`
`area 0.0.1.0`	`area 256`

5.3 Metric (Cost)

OSPF uses **cost** as its route metric. Every OSPF-enabled interface maintains its own interface cost.

Default formula:

$$InterfaceCost = \frac{100Mbit/s}{InterfaceBandwidth}$$

`100 Mbit/s` is OSPF's default reference bandwidth – **configurable**. **Route cost = accumulated cost** – the total of the outbound interface costs of every router the traffic passes through, from source to destination.

Recommendation: manually set interface cost based on actual bandwidth, rather than modifying the global OSPF reference bandwidth.

6 The Three OSPF Tables

OSPF maintains three distinct tables:

1. **OSPF Neighbor Table** – who am I talking to?
2. **LSDB** – what does the network look like? (raw topology, not routes)
3. **OSPF Routing Table** – what are my best paths? (this is the "protocol routing table" for OSPF specifically)

6.1 OSPF Neighbor Table

Before OSPF transmits link state information, **neighbor relationships must be established**, via exchanging **Hello packets**. Command: `display ospf peer`.

6.2 LSDB

Stores the LSAs generated locally *and* received from neighbors. **Type** indicates the LSA type; **AdvRouter** indicates the router that originated the LSA. Command: `display ospf lsdb`.

LSDB vs. OSPF Routing Table – the key distinction:

6.3 OSPF Routing Table

Contains **destination, cost, next hop** – the actual calculated routes (output of SPF). **Not the same as the router's global routing table (RIB)** – not all OSPF routes get added to it, since OSPF's routes must still compete against other protocols (by Preference, then Cost) at the local core routing table stage before reaching the RIB/FIB. Command: `display ospf routing`.

Route types shown: **Intra Area**, **Inter Area**, **ASE** (external, imported routes), **NSSA** (Not-So-Stubby Area routes).

7 OSPF Packet Format and Types

OSPF defines **5 types of packets**, sharing the same header format.

7.1 Encapsulation

OSPF packets are encapsulated **directly in IP packets** (not TCP or UDP). **IP protocol number: 89.**

7.2 The 5 OSPF Packet Types

Type	Packet Name	Function
1	Hello	Discovers and maintains neighbor relationships
2	Database Description (DD/DBD)	Exchanges brief/summary LSDB information between neighbors
3	Link State Request (LSR)	Requests specific detailed link state information
4	Link State Update (LSU)	Sends the actual detailed link state information (the real LSAs)
5	Link State Ack (LSAck)	Acknowledges receipt of LSAs

Common header fields: Version, Type, Packet Length, Router ID, Area ID, Checksum, Auth Type / Authentication.

7.3 Hello vs. DD vs. LSA

Packet	Purpose	Contains	Used in
Hello	Discover neighbors + keep relationship alive	Router ID, neighbor list, timers, DR/BDR info – no topology data	Down through Full, continuously
DD	Compare LSDB summaries	LSA headers only (Type, LS ID, AdvRouter, Sequence Number, Checksum) – like a table of contents	ExStart, Exchange
LSA (in LSU)	Deliver actual detailed link-state information	Full details: interface costs, connected neighbors, network segments	Loading (requested via LSR, delivered via LSU)

7.4 How the 5 Packet Types Map to the 4 Steps

Step	OSPF Packets Involved
1. Establish neighbor relationship	Hello packets
2. Exchange LSAs / sync LSDBs	DD → LSR → LSU → LSAck
3. Calculate optimal path (SPF)	Internal computation, no packet
4. Generate routing table	Internal computation, no packet

8 Neighbor Relationship Establishment

OSPF uses **Hello packets** to discover and establish neighbor relationships.

8.1 Transmission Mode

- On **Ethernet**, Hello packets are sent in **multicast** mode by default, to **224.0.0.5** – the reserved OSPF multicast address ("all OSPF routers").
- **224.0.0.6** is reserved for **DR/BDR** communication.
- On links without multicast support, OSPF falls back to **unicast**.

A Hello packet contains: the sender's **Router ID** and a **neighbor list** (router IDs already heard from).

8.2 The 3-Way Handshake (Down → Init → 2-Way)

State	Meaning
Down	Initial state – no packets received from this neighbor yet
Init	Received a Hello from the neighbor, but my own Router ID is not yet in their neighbor list
2-Way	Found my own Router ID listed in the neighbor's Hello packet – mutual visibility confirmed

Walkthrough (R1, R2):

1. R1 sends a Hello with an empty neighbor list. State: Down.
2. R2 receives it; R2's own ID is not listed → R2 = Init. R2 sends a new Hello, now listing R1.
3. R1 receives R2's Hello; finds its own ID listed → R1 = 2-Way.

Important: for two routers to reach 2-Way, certain Hello parameters **must match**: Hello Interval, Dead Interval, Network Mask (same segment), Area ID, and authentication settings. Mismatches are a very common real-world troubleshooting issue.

8.3 Purposes of Hello Packets

1. **Neighbor discovery** – automatically finding neighboring OSPF routers.
2. **Neighbor relationship establishment** – negotiating and matching parameters.

3. **Neighbor relationship holding (keepalive)** – periodic Hello exchange verifies neighbors are still alive.

Discovery vs. Establishment: discovery is simply noticing the neighbor exists (Down → Init → 2-Way); establishment is the broader process of negotiating and locking in a working relationship, which for DR/BDR-related pairs continues onward to a Full adjacency.

8.4 Key Fields in a Hello Packet

Field	Meaning
Network Mask	Subnet mask of the sending interface
Hello Interval	How often Hello packets are sent – default **10 seconds**
Router DeadInterval	If no Hello is received within this time, the neighbor is declared Down – default **40 seconds** (4× Hello Interval)
Neighbor	Router ID(s) already heard from
Router Priority	Used in DR/BDR election – default **1**; **0** means the router cannot participate
Designated Router	Interface address of the current DR
Backup Designated Router	Interface address of the current BDR

Options sub-fields: E (external routes supported), **MC** (multicast forwarding supported), **N/P** (whether the area is an NSSA).

9 Adjacency Establishment

9.1 States Beyond 2-Way

State	Meaning
ExStart	The router starts sending DD packets – no link state description yet, just negotiating Master/Slave
Exchange	Router and neighbor exchange DD packets containing LSDB **summaries**
Loading	Router and neighbor exchange LSR, LSU, and LSAck packets to fetch full LSA detail
Full	The router has **fully synchronized** its LSDB with the neighbor

Why go beyond 2-Way? 2-Way only confirms mutual Hello visibility – no topology data (LSAs) has been exchanged, so SPF cannot run yet. The whole point of ExStart→Full is to actually transfer LSAs between routers.

9.2 DD Packet Fields

Field	Meaning
I (Init)	1 if this is the first DD packet in a sequence; otherwise 0
M (More)	1 if more DD packets are coming; 0 if this is the last
MS (Master/Slave)	1 if the sender declares itself Master. The router with the **larger Router ID** becomes Master
DD Sequence Number	Ensures reliable, ordered, complete transmission between master and slave

Why elect a Master/Slave at all? OSPF runs directly over IP (protocol 89), not TCP – there is no built-in reliability or ordering. Electing one Master gives a single authoritative sequence-number timeline; the Slave simply follows and acknowledges each packet. This guarantees no ambiguity about ordering, no silently lost packets, and a clear signal (M=0) for when the exchange is complete.

9.3 Step-by-Step Walkthrough (R1 = 10.0.1.1, R2 = 10.0.2.2)

1. Both enter ExStart. R1 sends DD (Seq=X, I=1, M=1, MS=1). R2 sends DD (Seq=Y, I=1, M=1, MS=1). Both claim Master simultaneously.
2. R2's Router ID is larger → R2 becomes **Master**. R1 generates a Negotiation-Done event, moves ExStart → Exchange, and accepts the **Slave** role.
3. R1 (Slave) sends its LSDB summary: DD (Seq=Y, I=0, M=0, MS=0). R2 receives it and also moves to Exchange.
4. R2 (Master) sends its own LSDB summary: DD (Seq=Y+1, MS=1). R1 must acknowledge with an **empty** DD packet, same sequence number (Y+1).
5. After sending this acknowledgment, R1 generates an Exchange-Done event and moves to **Loading**. If R2's LSDB is already complete, R2 jumps straight to **Full**.

DD packet content: carries **LSA header information only** – LS Type, LS ID, Advertising Router, LS Sequence Number, LS Checksum (not full LSAs).

Other DD fields: Interface MTU – if the MTU in a received DD packet differs from the local MTU, the packet is discarded (MTU check is **disabled by default** on Huawei devices). **Options** – same meaning as in the Hello packet.

9.4 Loading: LSR / LSU / LSAck (Bidirectional)

Loading is **not** one-directional (Master → Slave only) – in the general case, **both** routers request whatever LSAs they are each missing, independent of their earlier Master/Slave roles (which only controlled the pace of the DD exchange).

1. R1 sends **LSR** packets requesting link-state info it discovered (via DD summaries) but doesn't have locally.
2. R2 responds with **LSU** packets containing the full, detailed LSA content requested.
3. Once R1 has everything it needs, R1 moves Loading → **Full**.
4. R1 sends an **LSAck** packet to acknowledge the LSU.

The same LSR/LSU/LSAck flow happens in the opposite direction if R2 is also missing information from R1. Both sides reach Full once each has received everything it requested.

10 DR and BDR

10.1 The Problem on Multi-Access (MA) Networks

If every router on a shared segment formed a full adjacency with every other router:

- **n(n-1)/2 adjacencies** would be needed – grows quadratically with the number of routers.
- This **complicates management**.
- **Repeated LSA flooding wastes resources**.

10.2 The Solution: DR and BDR

- The **DR (Designated Router)** establishes and maintains adjacencies with **all other routers** on the segment, and handles synchronizing LSAs for the whole segment. Other routers do not directly exchange link-state information with each other.
- The **BDR (Backup Designated Router)** exists to prevent a **Single Point of Failure (SPOF)** – ready to immediately take over if the DR fails.
- **DRother** = any router that is neither DR nor BDR.

10.3 Multi-Access Network Types

Type	Description / Example
BMA (Broadcast Multi-Access)	Multiple routers share a broadcast-capable medium – **Ethernet**
NBMA (Non-Broadcast Multi-Access)	Multiple routers share a medium without broadcast support – **Frame Relay**

10.4 DR and BDR Election Rules

1. **Non-preemptive** – once elected, a new router joining later with a higher priority/ID does *not* take over.
2. Election is **per-interface**, not per-router.
3. Higher **DR priority** on the interface wins.
4. Tiebreaker: larger **Router ID** wins.
5. **Priority = 0** means that interface does not participate at all.

If all routers have priority 0: no DR/BDR is ever elected. OSPF still works – routes are still calculated correctly – but every router must form a full adjacency with every other router directly, losing the efficiency benefit of having a DR.

10.5 Election Process

1. When an interface comes Up, it sends Hello packets and enters a **Waiting** state, governed by a **Waiting Timer** equal to the Dead Timer – default **40 seconds**, not configurable.
2. Before the Waiting Timer expires, Hello packets carry no DR/BDR field.
3. If, during the wait, a received Hello **already** contains a DR/BDR field, no election is triggered – the router accepts the existing DR/BDR and moves directly to neighbor synchronization.
4. Any newly connected router **accepts the existing DR and BDR regardless of its own Router ID or priority** (non-preemption in action).
5. If the DR fails: the **BDR is promoted to DR immediately**; remaining routers with priority > 0 then elect a **new BDR**.

10.6 Full Adjacency Scope

Full adjacencies form between:

- DR $\leftrightarrow$ every DRother
- BDR $\leftrightarrow$ every DRother
- DR $\leftrightarrow$ BDR

DRother $\leftrightarrow$ DRother pairs remain at 2-Way – they never form a direct Full adjacency, since both are synchronized via the DR anyway.

10.7 DR/BDR Election by OSPF Network Type

OSPF Network Type	Common Data-Link Protocol	Elects DR?	Adjacency Behavior
Point-to-point (P2P)	PPP, HDLC	No	Direct adjacency – only 2 routers on the link
Broadcast	Ethernet	Yes	DR $\leftrightarrow$ BDR $\leftrightarrow$ DRothers, as above
NBMA	Frame Relay	Yes	Same DR/BDR logic, manually configured neighbors
P2MP	Manually specified	No	No DR/BDR – treated like multiple P2P links

10.8 Adjusting the Network Type

OSPF sets the network type **automatically** based on the interface's data-link encapsulation – Ethernet defaults to **Broadcast**, triggering DR/BDR election **even on links that are logically point-to-point** (e.g. a single Ethernet cable directly connecting two routers).

This is often **unnecessary and time-consuming** – the DR/BDR election's 40-second Waiting Timer delays reaching Full. Since the link is logically a P2P connection, DR/BDR election is not needed at all.

Fix: manually change the network type to **P2P**, skipping election entirely and reaching ExStart directly – speeding up adjacency establishment significantly.

11 OSPF Configuration Commands

11.1 Basic Setup

1. **Create an OSPF process and enter OSPF view:**

   `[Huawei] ospf [process-id | router-id router-id]`

   OSPF supports multi-process; the process ID is locally significant. **Two devices using different OSPF process IDs can still establish an adjacency.**

2. **Create an OSPF area and enter area view:**

   `[Huawei-ospf-1] area area-id`

3. **Enable OSPF on a network segment (area view):**

   `[Huawei-ospf-1-area-0.0.0.0] network network-address wildcard-mask`

   The interface's IP mask length must be $\geq$ the mask length specified here, and its primary IP must fall within the specified segment.

4. **Enable OSPF directly on an interface (alternative):**

   `[Huawei-GigabitEthernet1/0/0] ospf enable process-id area area-id`

   `ospf enable` **takes precedence** over the `network` command. By default, OSPF advertises a loopback's IP as a /32 host route regardless of its configured mask – to advertise the loopback's actual network segment, set its network type to NBMA or broadcast.

11.2 Interface Tuning

5. **Set DR election priority:**

   `[Huawei-GigabitEthernet1/0/0] ospf dr-priority priority`

   Default: **1**.

6. **Set Hello interval:**

   `[Huawei-GigabitEthernet1/0/0] ospf timer hello interval`

   Default: **10 seconds** (P2P or broadcast). Dead interval defaults to **4**$\times$ the Hello interval.

7. **Set OSPF network type:**

   `[Huawei-GigabitEthernet1/0/0] ospf network-type {broadcast | nbma | p2mp | p2p}`

   Default is auto-determined: **Ethernet** $\rightarrow$ **Broadcast**; **Serial/POS (PPP or HDLC)** $\rightarrow$ **P2P**.

11.3 Configuration Example

Five routers, all in Area 0. Naming convention: Router ID `10.0.x.x` (x = router number); interconnection IPs `10.0.xyz.x/24` where xyz = router numbers in ascending order.

R2's configuration:

```
[R2] ospf 1 router-id 10.0.2.2
[R2-ospf-1] area 0.0.0.0
[R2-ospf-1-area-0.0.0.0] network 10.0.12.0 0.0.0.255
[R2-ospf-1-area-0.0.0.0] network 10.0.24.2 0.0.0.0
[R2-ospf-1-area-0.0.0.0] network 10.0.235.2 0.0.0.0
```

Wildcard mask note: `0.0.0.255` covers a whole /24 segment; `0.0.0.0` matches only that exact single IP – useful for enabling OSPF on one specific interface address without including the whole subnet.

12 OSPF Verification Commands

Command	Shows
`display ospf interface all`	Per-interface OSPF settings: cost, type, MTU, priority, state, DR/BDR addresses, timers
`display ospf peer`	**Neighbor Table** – relationship status, Master/Slave mode, DR/BDR
`display ospf lsdb`	**LSDB** – raw topology/link facts (Type, LinkState ID, AdvRouter, Age, Sequence, Metric)
`display ospf routing`	**OSPF Routing Table** – calculated routes (Destination, Cost, Type, NextHop, AdvRouter, Area)

12.1 Key Verification Insights

- On a **P2P** interface (e.g. Serial, PPP/HDLC), `display ospf interface all` shows **State: P-2-P** with **no DR/BDR fields** – confirming P2P links skip DR/BDR election entirely.
- Correspondingly, `display ospf peer` on that same P2P link shows **DR: None, BDR: None** – the clearest real-world confirmation that P2P links don't elect a DR/BDR.
- On a **Broadcast** (Ethernet) interface, the same commands show a real DR/BDR address and a **State** of DR, BDR, or DROther for the local router.
- `display ospf routing` on a router shows it has learned routes to the **entire network** through OSPF – confirming the LSDB → SPF → routing table pipeline works end-to-end.